

NPA Hierarchy for Quantum Isomorphism and Homomorphism Indistinguishability

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Joint work with Prem Nigam Kar, David E. Roberson, and Tim Seppelt
*NPA Hierarchy for Quantum Isomorphism and
Homomorphism Indistinguishability* [arXiv:2407.10635](https://arxiv.org/abs/2407.10635)

¹ERC, POCOCOP, 101071674

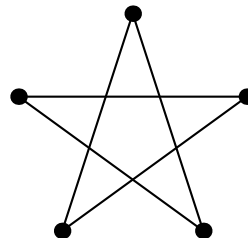
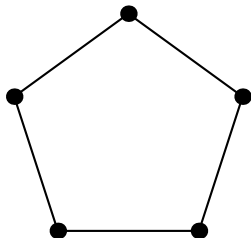
Graph isomorphisms and homomorphisms

Definition. A map $\varphi: V(G) \rightarrow V(H)$ is an **isomorphism** if $\varphi(u)\varphi(v) \in E(H) \Leftrightarrow uv \in E(G)$. We write $G \cong H$.

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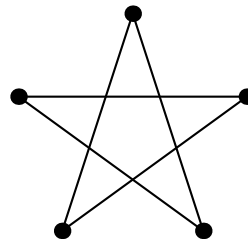
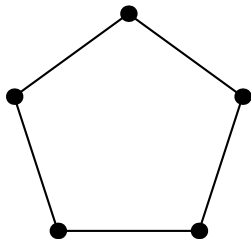
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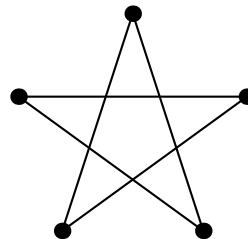
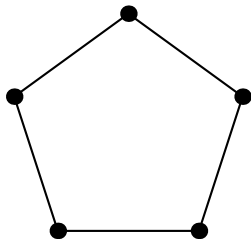


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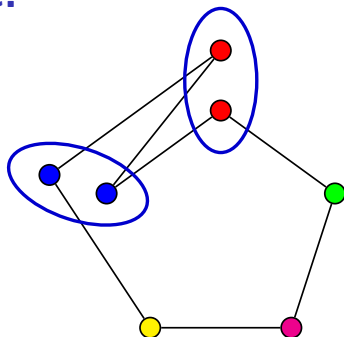
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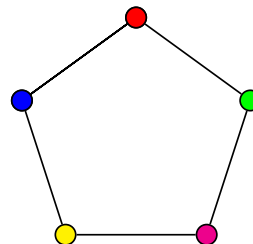


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$C_7 \rightarrow C_5$



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The class $\mathcal{F}$	The relation $G \cong_{\mathcal{F}} H$
All graphs	Isomorphism [Lovász 1967]
Cycles	Cospectrality
Cycles & paths	Cospectral & cospectral complements
Trees	Fractional isomorphism [Dvořák 2010]
Treewidth $\leq k$	Indistinguishable by k -WL [Dvořák 2010]
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Benefits. (1) randomized poly-time algorithm for k^{th} level,
(2) more elementary proof avoiding quantum groups.

Main theorem

The k^{th} level of the NPA hierarchy for the (G, H) -isomorphism game is feasible



There exists a level- k quantum isomorphism map from G to H



G and H are homomorphism indistinguishable over $\mathcal{P}_k$

Nonlocal games (Bell games, Bell inequalities)

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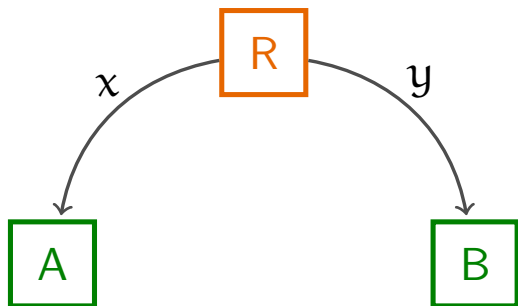
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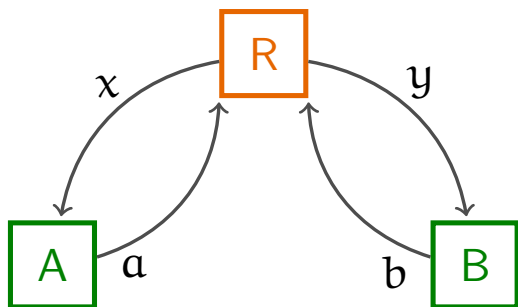
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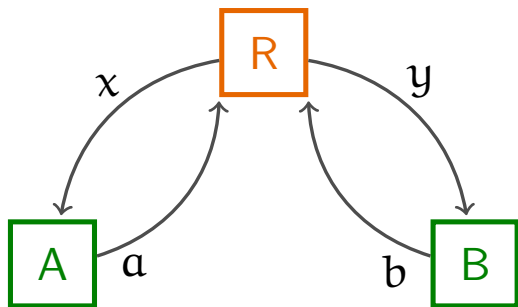
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- Players win if $V(a, b|x, y) = 1$.

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Example: Clauser, Horne, Shimony, Holt (CHSH) game

- $X = Y = A = B = \{0, 1\}$,
- π is uniform,
- $V(a, b|x, y) = \begin{cases} 1 & \text{if } a \oplus b = x \wedge y, \\ 0 & \text{if } a \oplus b \neq x \wedge y. \end{cases}$

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Quantum. Players share a unit vector $|\psi\rangle \in \mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.

- A POVM $\mathcal{E}_x = \{E_x^a \in \mathbb{C}^{d_A \times d_A} : a \in A\}$, for each $x \in X$.
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The **quantum value** $\omega^*(\mathcal{G})$ of a game $\mathcal{G}$ is the supremum of

$$\sum_{x,y} \pi(x,y) \sum_{a,b} V(a, b | x, y) p(a, b | x, y).$$

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Thm. $\omega(\text{CHSH}) = 3/4 < \cos^2(\pi/8) = \omega^*(\text{CHSH})$.

(G, H) -Isomorphism Game

Intuition: Alice and Bob want to convince a referee that $G \cong H$.

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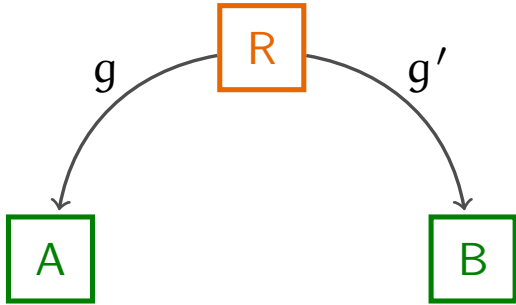
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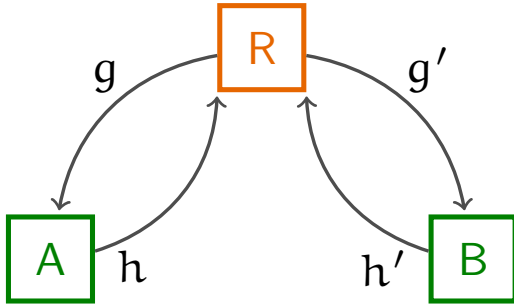


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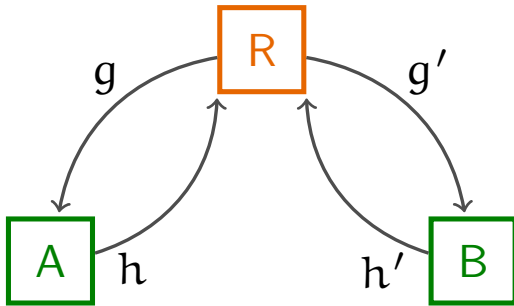


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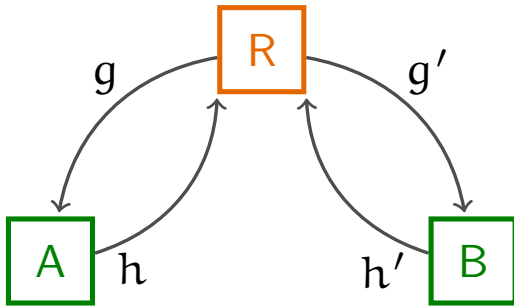
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where rel denotes how two vertices are “related”.

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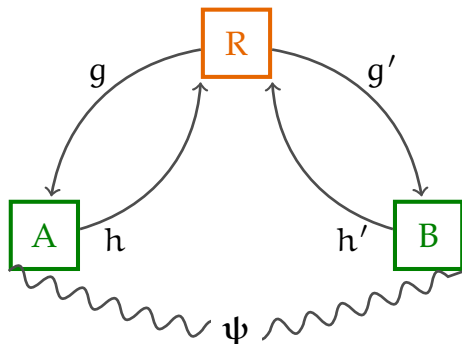
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Proposition. $G \cong H \Leftrightarrow$ Classical players can **win** the game.

Quantum isomorphism

$G \cong_{qc} H :=$ **Quantum** players can win the (G, H) -isomorphism game.

Quantum strategies

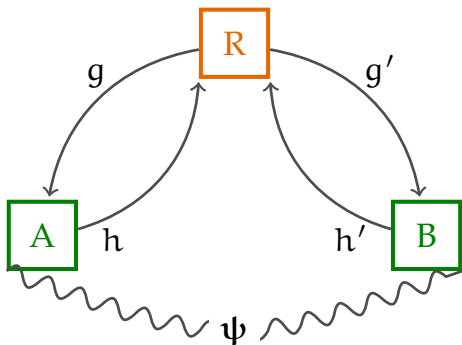


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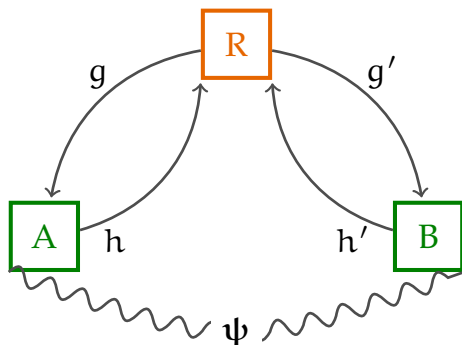
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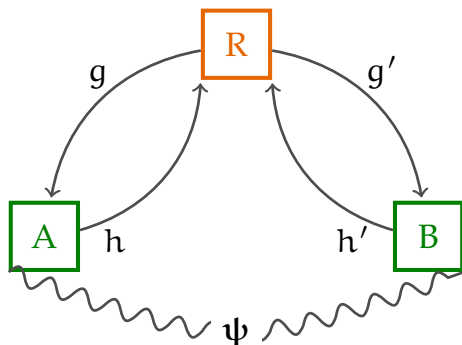


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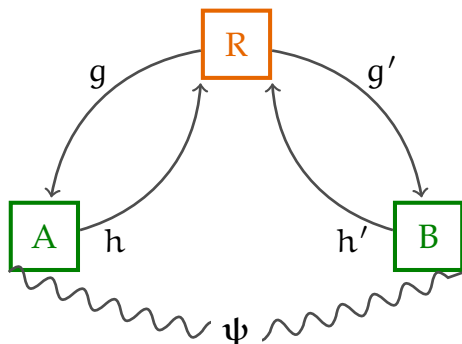


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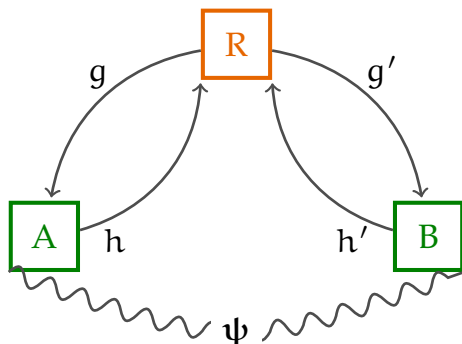


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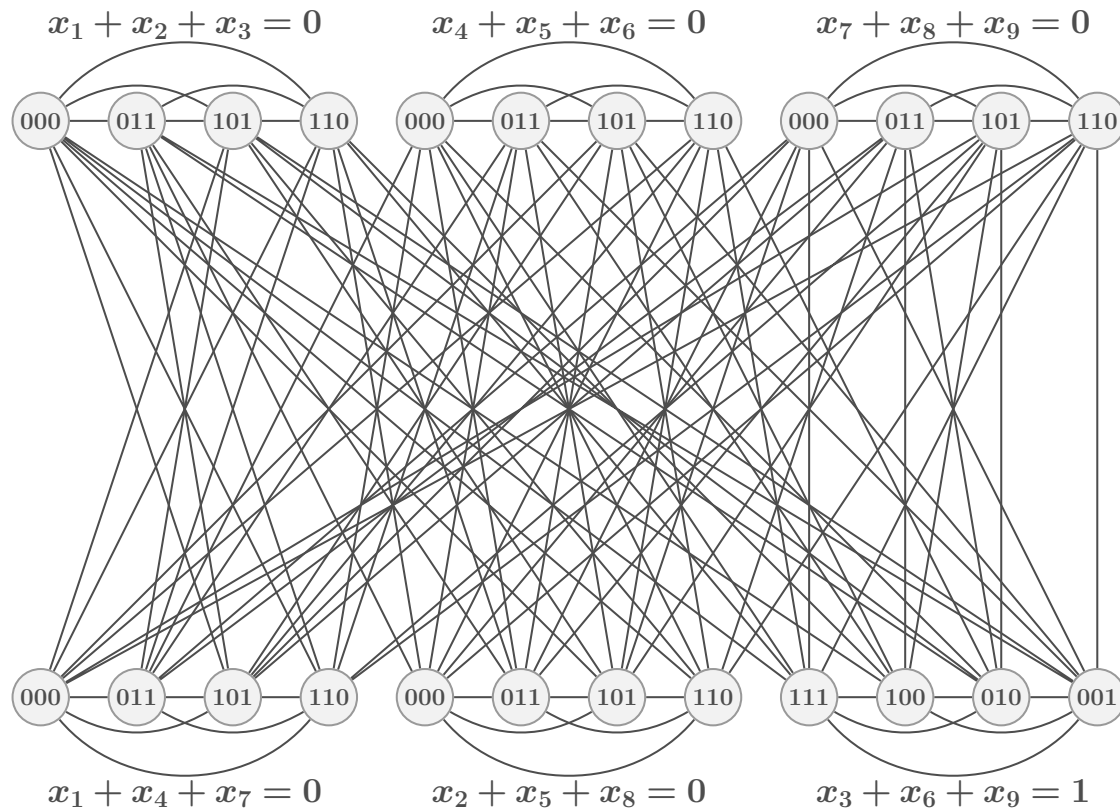


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The probability that players respond with h, h' on questions g, g' is

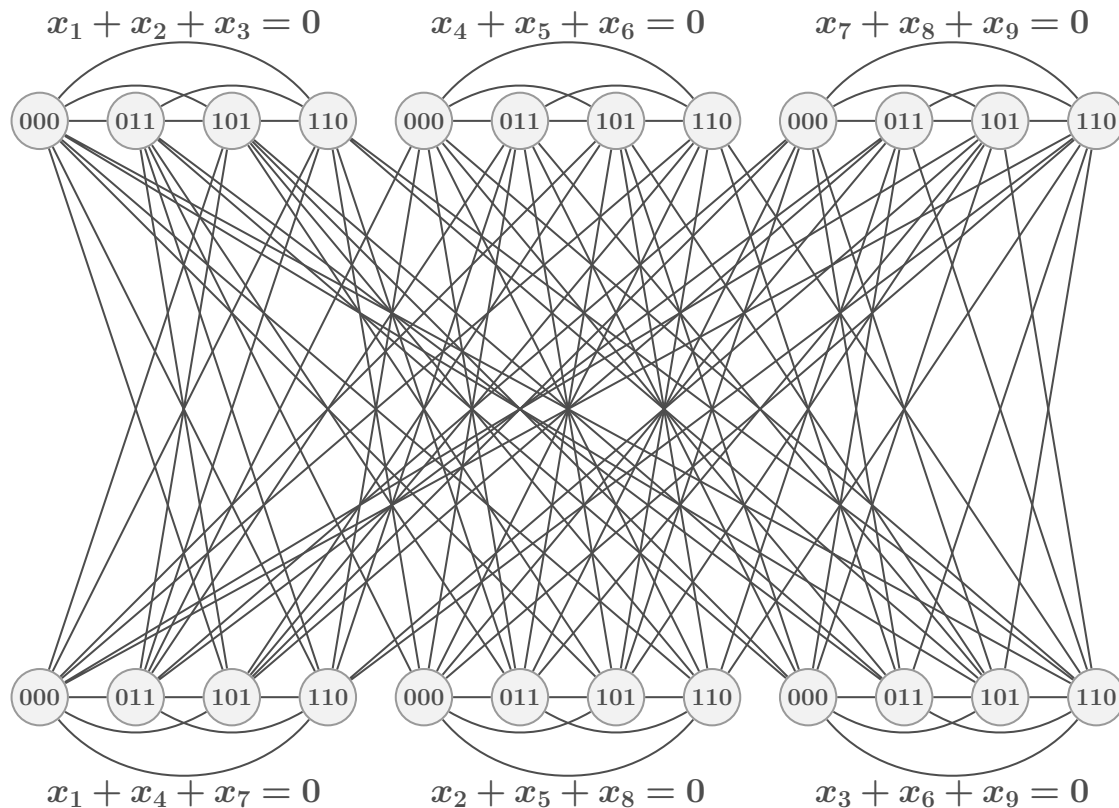
$$p(h, h'|g, g') = \langle \psi, E_{gh} F_{g'h'} \psi \rangle$$

Example: $G \not\cong H$ but $G \cong_{qc} H$



Construction based on reduction from linear system games.

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Quantum permutation matrix

A matrix $P = (p_{ij})$ is **quantum permutation matrix** if p_{ij} are elements of an C^* -algebra s.t.

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Thm (Lupini, Mančinska, Roberson). $G \cong_{qc} H \Leftrightarrow A_G P = P A_H$, for some quantum permutation matrix P . This is similar to

$$\begin{aligned} G \cong H &\Leftrightarrow A_G P = P A_H && \text{(isomorphism),} \\ G \cong_{\mathcal{T}} H &\Leftrightarrow A_G D = D A_H && \text{(fractional isomorphism).} \end{aligned}$$

Complexity theory

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$$\text{MIP}^* = \text{RE}$$

(Ji, Natarajan, Vidick, Wright, Yuen, 2022)

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MIP: several non-communicating provers convince the verifier.

MIP*: the provers may additionally share quantum entanglement.

Bilabelled graphs

Definition.

A (k, k) -**bilabelled graph** is a triple $\mathbf{F} = (F, \mathbf{u}, \mathbf{v})$ where

- F is a graph;
- $\mathbf{u} = (u_1, \dots, u_k)$, $\mathbf{v} = (v_1, \dots, v_k) \in V(F)^k$.

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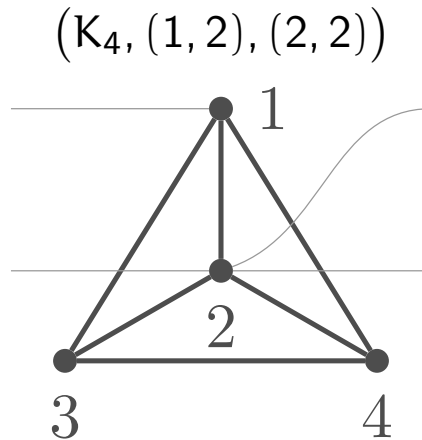
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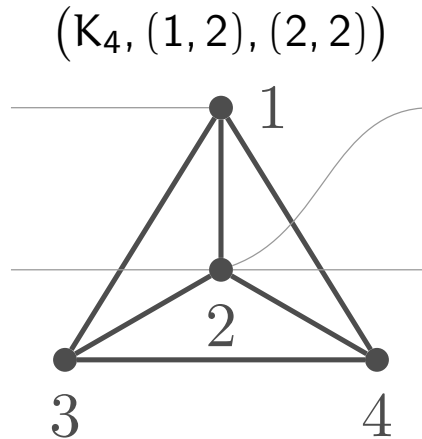
Example. $F = (K_4, (1, 2), (2, 2))$.

How to draw bilabelled graphs

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A bilabelled graph is **planar** if it can be drawn with no crossings.

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So $\mathbf{A}_G$ is the **adjacency matrix** of G .

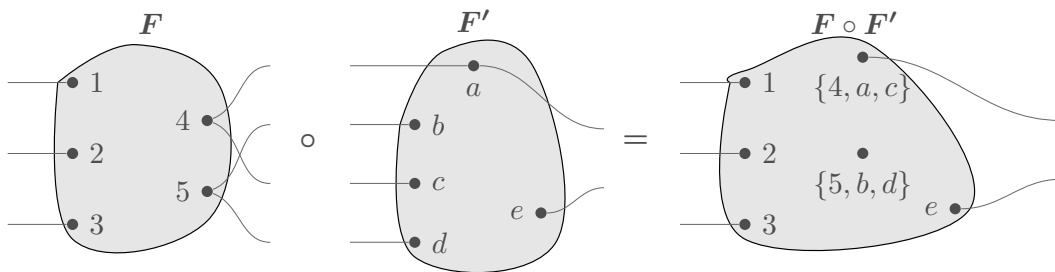
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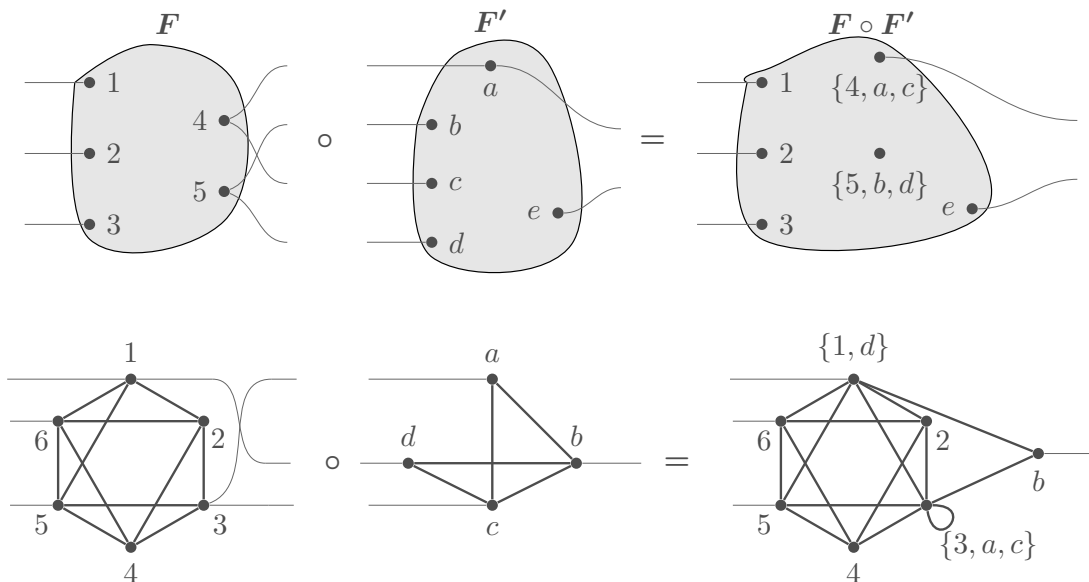


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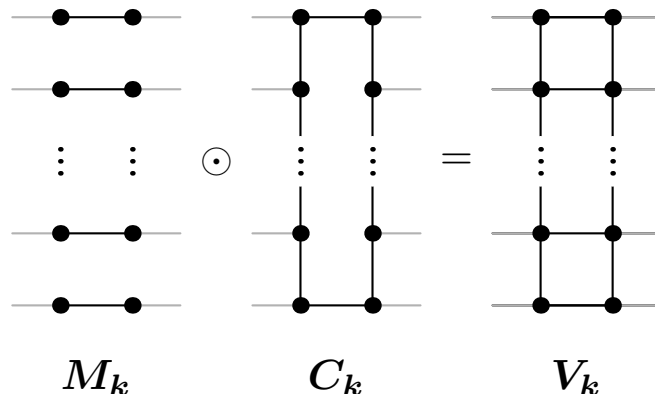
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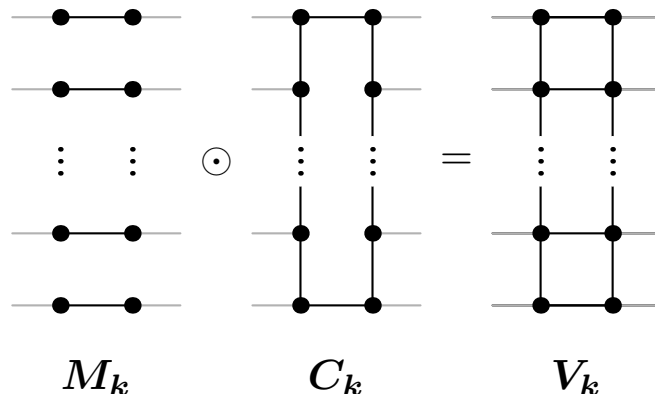


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Other operations: transposition and cyclic permutations.

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- The other direction works too.

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Let $k \in \mathbb{N}$. For all $\ell \leq k$, define

$$|\psi_{g_1 h_1 \dots g_\ell h_\ell}\rangle := E_{g_1 h_1} E_{g_2 h_2} \dots E_{g_\ell h_\ell} |\psi\rangle$$

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This gives rise to a semidefinite program $\rightarrow$ NPA hierarchy.

NPA hierarchy for the isomorphism game

Let $\Sigma = V(G) \times V(H)$.

A matrix $\mathcal{R} \in \mathbb{M}_{\Sigma^{\leq k}}(\mathbb{C})$ is a **certificate for the k^{th} level of the NPA hierarchy for the (G, H) -isomorphism game** if

- ① $\mathcal{R} \succeq 0$,
- ② $\mathcal{R}_{\varepsilon, \varepsilon} = 1$,
- ③ $\mathcal{R}_{s, t}$ depends only on the equivalence class of $s^R t$,
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Thm. (G, H) -isomorphism game has a perfect quantum strategy iff there is a certificate for the k^{th} level of the NPA hierarchy, $k \in \mathbb{N}$.

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Thm (Choi). Φ is completely positive $\Leftrightarrow C_\Phi$ is positive.

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- 4 The constraints on $\mathcal{R}$ translate into constraints on the map Φ .

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- ① Φ is completely positive;
- ② $\Phi(I) = I = \Phi^*(I)$; (thus Φ is **unital** and **trace-preserving**)
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Converse requires some previous results and a bit of combinatorics.

Recap

The k^{th} level of the NPA hierarchy for the (G, H) -isomorphism game is feasible



There exists a level- k quantum isomorphism map from G to H



G and H are homomorphism indistinguishable over $\mathcal{P}_k$

Corollary

Corollary. $G \cong_{qc} H$ if and only if $G \cong_{\mathcal{P}} H$, where $\mathcal{P} = \cup_{k=1}^{\infty} \mathcal{P}_k$.

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All that is left is to show that $\mathcal{P}$ is the class of all planar graphs.

Proof that $\mathcal{P} \subseteq \text{planar graphs}$: All generators of $\mathcal{P}_k$ are planar, all operations preserve planarity.

Proof that $\mathcal{P} \supseteq$ planar graphs

Lemma. Each $\mathcal{P}_k$ is minor-closed, and thus so is each $\mathcal{P}_k$ and $\mathcal{P}$.

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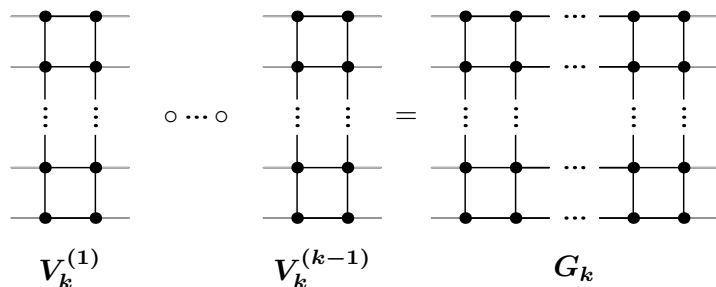
Lemma. The class $\mathcal{P}_k$ contains the $k \times k$ grid.

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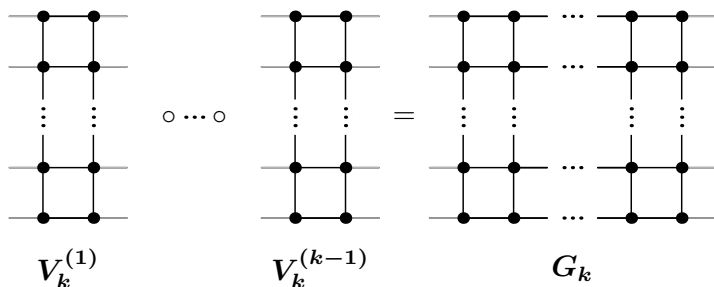


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Well-known: Every planar graph is the minor of some $k \times k$ grid.

Some remarks/questions

- $\mathcal{P}_k$ has treewidth bounded by $3k - 1$. This implies there is a randomized poly time algorithm for determining if $G \cong_{\mathcal{P}_k} H$, and thus whether the k^{th} level of the NPA hierarchy for the (G, H) -isomorphism game is feasible.

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- There are graphs G and H of size $\leq 72k^2$ that are not quantum isomorphic, but the k^{th} level of the NPA hierarchy is feasible for the (G, H) -isomorphism game.
- Can we obtain a better description of the classes $\mathcal{P}_k$?

Thank you!

The hierarchy of Navascués, Pironio, and Acín (NPA)

CHSH game, $X = Y = A = B = \{0, 1\}$. Deterministic strategies M :

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Representing the CHSH game as an operator K yields

$$K = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & \\ & & & 1 \\ & & & 1 \\ 1 & 1 & 1 & \end{pmatrix}$$

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The classical value of the game is $3/4$.

Nonlocal games

Def. A **nonlocal game** is a 6-tuple (X, Y, A, B, π, V) , where

- ① X, Y, A , and B are finite and nonempty sets,
- ② $\pi \in P(X \times Y)$ is a probability vector, and
- ③ $V: A \times B \times X \times Y \rightarrow \{0, 1\}$ is a predicate.

The sets X, Y are questions and A, B are answers. The predicate $V(a, b|x, y)$ determines whether the players win or lose.

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We can think of **strategies** as being represented by operators

$$M \in L(\mathbb{R}^X \otimes \mathbb{R}^Y, \mathbb{R}^A \otimes \mathbb{R}^B).$$

The value $M(a, b|x, y)$ represents the probability that Alice and Bob answer (x, y) with (a, b) .

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The probability that M wins a game is

$$\sum_{(x,y) \in X \times Y} \pi(x, y) \sum_{(a,b) \in A \times B} V(a, b|x, y) M(a, b|x, y) = \langle K, M \rangle,$$

where $K(a, b|x, y) = \pi(x, y)V(a, b|x, y)$.

Commuting measurement strategies

An operator M represents a **commuting measurement strategy** if there exists a Hilbert space $\mathcal{H}$, a unit vector $u \in \mathcal{H}$, and projection operators

$$\{P_a^x : x \in X, a \in A\} \quad \text{and} \quad \{Q_b^y : y \in Y, b \in B\}$$

acting on $\mathcal{H}$ such that the following are satisfied:

- ① $\sum_{a \in A} P_a^x = 1_{\mathcal{H}}$ and $\sum_{b \in B} Q_b^y = 1_{\mathcal{H}}$, $x \in X, y \in Y$,
- ② $[P_a^x, Q_b^y] = 0$, $x \in X, y \in Y, a \in A, b \in B$,
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A **commuting measurement value** of game G is

$$\omega^c(G) = \sup_{M \in \mathcal{C}} \langle K, M \rangle,$$

where K is defined from G as before and $\mathcal{C}$ is the class of commuting measurement strategies.

1st level of the NPA hierarchy (intuition)

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- ③ $R((x, a), (x, c)) = 0$, $x \in X, a, c \in A, a \neq c$,
 $R((y, b), (y, d)) = 0$, $y \in Y, b, d \in B, b \neq d$,
(**P_a^x and P_c^x , Q_b^y and Q_d^y are orthogonal**)

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- ② $\sum_{a \in A} R((x, a), s) = R(\varepsilon, s)$ and $\sum_{a \in A} R(s, (x, a)) = R(s, \varepsilon)$,
 $\sum_{b \in B} R((y, b), s) = R(\varepsilon, s)$ and $\sum_{b \in B} R(s, (y, b)) = R(s, \varepsilon)$,
(**summing over operators in measurements is identity**)
- ③ $R((x, a), (x, c)) = 0$, $x \in X, a, c \in A, a \neq c$,
 $R((y, b), (y, d)) = 0$, $y \in Y, b, d \in B, b \neq d$,
(**P_a^x and P_c^x , Q_b^y and Q_d^y are orthogonal**)
- ④ $R((z, c), (z, c)) = R(\varepsilon, (z, c)) = R((z, c), \varepsilon)$, (**$P^2 = P$**)

1st level of the NPA hierarchy (intuition)

For a commuting measurement strategy M , we would like to capture the numbers $M(a, b|x, y) = \langle u, P_a^x Q_b^y u \rangle$.

We can consider the **Gram matrix** of the vectors:

$$\{u\} \cup \{P_a^x u : x \in X, a \in A\} \cup \{Q_b^y u : y \in Y, b \in B\}.$$

Let $\Sigma^{\leq 1} = (X \times A) \sqcup (Y \times B) \cup \{\varepsilon\}$.

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- ④ $R((z, c), (z, c)) = R(\varepsilon, (z, c)) = R((z, c), \varepsilon)$, (**$P^2 = P$**)
- ⑤ $R((x, a), (y, b)) = R((y, b), (x, a))$. (**commutativity**)

1st level of the NPA hierarchy (intuition)

Let $\mathcal{C}_1$ be the class containing all strategies M for which there is a positive semidefinite R such that $M(a, b|x, y) = R((x, a), (y, b))$.

We have

$$\omega^c(G) = \sup_{M \in \mathcal{C}} \langle K, M \rangle \leq \sup_{M \in \mathcal{C}_1} \langle K, M \rangle.$$

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If we define a Hermitian operator $H \in L(\mathbb{C}^{\Sigma^{\leq 1}}, \mathbb{C}^{\Sigma^{\leq 1}})$ by

$$H((x, a), (y, b)) = H((y, b), (x, a)) = \frac{1}{2} \pi(x, y) V(a, b|x, y),$$

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$$\langle K, M \rangle = \langle H, R \rangle,$$

which gives us a semidefinite program, where we optimize $\langle H, R \rangle$ over positive semidefinite R satisfying (affine) linear constraints given in items 1–5 above.

k^{th} level of the NPA hierarchy (intuition)

In the k^{th} level of the NPA hierarchy we consider operators R indexed by $\Sigma^{\leq k}$ satisfying conditions similar to 1–5.

Then the class $\mathcal{C}_k$ contains all strategies M for which there exists such admissible operator R . We have:

$$\mathcal{C}_1 \supseteq \mathcal{C}_2 \supseteq \mathcal{C}_3 \supseteq \cdots \supseteq \mathcal{C}.$$

Thm. The following are equivalent:

- M is a commuting measurement strategy.
- $M \in \mathcal{C}_k$ for every k .

Equivalently:

$$\mathcal{C} = \bigcap_{k=1}^{\infty} \mathcal{C}_k.$$